

Interview Questions - Python Basics

Basic Python Concept Questions

1. **Reverse a String:** Write a function to reverse a string without using any built-in reverse methods or slicing with negative indices.
2. **Reverse a Number:** Given an integer (e.g., 12345), reverse its digits to get 54321 without converting it to a string.
3. **Find Maximum in a List:** Find the largest number in a list without using the `max()` function or any sorting methods.
4. **Find Second Maximum:** Find the second largest number in a list without using built-in functions like `max()` or `sorted()`.
5. **Check Palindrome:** Determine if a given string is a palindrome (reads the same forwards and backwards) without using string reversal methods.
6. **Find Duplicates:** Given a list of numbers, find and return all duplicate elements without using sets or built-in counting functions.
7. **Count Vowels:** Count the number of vowels (a, e, i, o, u) in a given sentence, handling both uppercase and lowercase letters.
8. **Word Frequency Counter:** Given a sentence, count the frequency of each unique word without using `Counter` from collections.
9. **Check Prime Number:** Write a function to check if a given number is prime without using any mathematical libraries.
10. **Fibonacci Sequence:** Generate the first n numbers in the Fibonacci sequence without using recursion.
11. **Remove Duplicates from List:** Remove all duplicate elements from a list while maintaining the original order, without using sets.
12. **Find Missing Number:** Given a list containing numbers from 1 to n with one number missing, find the missing number without sorting.
13. **Anagram Check:** Check if two strings are anagrams of each other (contain the same characters in different order) without using `sorted()` or `Counter`.
14. **Sum of Digits:** Given an integer, find the sum of its digits without converting it to a string.
15. **Count Consonants:** Count the number of consonants in a given string (all alphabetic characters that are not vowels), ignoring spaces and special characters.

16. **Factorial Calculation:** Calculate the factorial of a number (e.g., $5! = 5 \times 4 \times 3 \times 2 \times 1$) without using any built-in factorial functions or recursion.
17. **Find Common Elements:** Given two lists, find all common elements between them without using set operations like intersection.
18. **Check Armstrong Number:** Determine if a number is an Armstrong number (sum of cubes of its digits equals the number itself, e.g., $153 = 1^3 + 5^3 + 3^3$) without string conversion.
19. **Merge Two Sorted Lists:** Merge two sorted lists into one sorted list without using the `sort()` or `sorted()` functions.
20. **Count Specific Character:** Count how many times a specific character appears in a string without using the `count()` method.
21. **Find All Prime Numbers in Range:** Generate all prime numbers between 1 and n without using any mathematical libraries or pre-built functions.
22. **Swap Variables:** Swap two variables without using a third temporary variable or tuple unpacking.
23. **Find GCD (Greatest Common Divisor):** Find the GCD of two numbers without using the `math.gcd()` function or any built-in methods.
24. **Remove Specific Element:** Remove all occurrences of a specific element from a list without using `remove()`, `filter()`, or list comprehensions.
25. **Check if List is Sorted:** Determine whether a given list is sorted in ascending order without using any sorting or comparison functions beyond basic operators.